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Third-generation Drones

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4.1 Introduction

The third generation of drones, emerging from the early 2000s to the mid2010s, represents a pivotal era in the evolution of unmanned aerial vehicle (UAV) technology. This generation marked the transition from semi-autonomous, reconnaissance-focused systems to intelligent, networked, and multirole UAV platforms capable of performing complex missions across military, scientific, and civilian domains. Building on the technological foundations of the second generation particularly advancements in GPS navigation, digital telemetry, and real-time imaging third-generation drones integrated breakthroughs in miniaturized electronics, advanced sensors, and data fusion technologies, enabling enhanced autonomy, endurance, and situational awareness.

This period witnessed the rapid rise of medium-altitude long-endurance (MALE) and high-altitude long-endurance (HALE) UAVs, such as the MQ-1 Predator, RQ-4 Global Hawk, and Heron, which redefined the strategic role of drones in surveillance, target acquisition, and precision strike missions. The integration of satellite communication (SATCOM) links allowed these drones to operate beyond visual and radio line-of-sight, providing real-time data streams to command centers thousands of kilometers away. The inclusion of advanced autopilot systems, electro-